35) In a reaction $x + y + z_2 \rightarrow xyz_2$ identify the Limiting reagent if any, in the following reaction mixtures.

(a) 200 atoms of x + 200 atoms of y + 50 molecules of z_2

(b) 1mol of x + 1 mol of y+3 mol of z_2

(c) 50 atoms of x + 25 atoms of y+50 molecules of z_2

d) 2.5 mol of x +5 mol of y+5 mol of z_2

36) Mass of one atom of an element is 6.645×10^{-23} g. How many moles of element are there in 0.320 kg.

37) What is the difference between molecular mass and molar mass? Calculate the molecular mass and molar mass for carbon monoxide.

38) What is the empirical formula of the following ?

i) Fructose ($C_6H_{12}O_6$) found in honey

ii) Caffeine ($C_8H_{10}N_4O_2$) a substance found in tea and coffee.

39) The reaction between aluminium and ferric oxide can generate temperatures up to 3273 K and is used in welding metals. (Atomic mass of Al = 27 u Atomic mass of O = 16 u)

$2Al + Fe_2O_3 \rightarrow Al_2O_3 + 2Fe$; If, in this process, 324 g of aluminium is allowed to react with 1.12 kg of ferric oxide.

i) Calculate the mass of Al_2O_3 formed.

ii) How much of the excess reagent is left at the end of the reaction?

40) How many moles of ethane is required to produce 44 g of $CO_{2\,(g)}$ after combustion.

41) Hydrogen peroxide is an oxidising agent. It oxidises ferrous ion to ferric ion and reduced itself to water. Write a balanced equation.

42) Calculate the empirical and molecular formula of a compound containing 76.6% carbon, 6.38 % hydrogen and rest oxygen its vapour density is 47.

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